Indian Institute of Technology Madras

Department of Mathematics

AgroAI: An Integrated ML & DL-Based System for Crop Prediction, Health Monitoring, and Smart Irrigation Using Soil and Weather Analytics

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Chapter 1

Introduction

1.1 Background

Agriculture is a critical sector that directly impacts food security, economic stability, and sustainable development, especially in agrarian economies. However, modern agriculture faces numerous challenges, including unpredictable climate conditions, soil degradation, water scarcity, and pest and disease outbreaks. To address these challenges and boost productivity, the integration of advanced technologies such as Machine Learning (ML) and Deep Learning (DL) has become increasingly essential.

AgroAI is a comprehensive AI-powered solution designed to revolutionize traditional farming practices through data-driven decision-making. The system integrates soil properties, weather analytics, and historical crop data to provide intelligent support in three core areas:

- **Crop Recommendation**: Leveraging ML models, this module analyzes soil characteristics (such as pH, nitrogen, phosphorus, and potassium levels), weather conditions (temperature, humidity, rainfall), and past crop trends to recommend the most suitable crop for a given region. This helps optimize yield and ensures sustainable use of land resources.

- **Disease Detection**: Using DL techniques, particularly convolutional neural networks (CNNs), the system identifies plant diseases from leaf images and provides targeted recommendations for treatment. This module enhances early detection and prevention, reducing losses due to pests and infections.

- **Smart Irrigation**: This module utilizes real-time soil moisture, temperature, and weather forecasts to optimize water usage. ML algorithms help determine the right amount of irrigation needed at specific times, thereby conserving water and improving plant health.

  By combining these three components, AgroAI aims to empower farmers with intelligent tools that enhance productivity, reduce environmental impact, and contribute to the broader goals of precision agriculture and food sustainability.

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1.2 Problem statement

Traditional farming practices often rely on intuition and past experience, which can lead to suboptimal decisions regarding crop selection, disease management, and irrigation scheduling. Farmers face critical challenges such as inaccurate crop planning, delayed detection of plant diseases, and inefficient water usage—all of which contribute to reduced yield, increased costs, and environmental stress. Moreover, the lack of timely, data-driven insights limits their ability to adapt to dynamic soil and weather conditions.

There is a pressing need for an integrated, intelligent system that can assist farmers in making informed decisions by leveraging modern technologies. The solution should provide personalized crop recommendations, early disease detection with treatment suggestions, and real-time irrigation management—based on soil health metrics, weather data, and historical agricultural trends.

AgroAI aims to address this gap by developing a unified Machine Learning and Deep Learning-based system that offers:

Smart crop recommendation based on soil nutrients and climatic factors,

Automated plant disease recognition using image analytics and DL models, and

Intelligent irrigation scheduling to optimize water use based on environmental and soil moisture data.

This system is intended to support sustainable agriculture, increase crop productivity, and empower farmers with actionable insights using accessible and scalable AI technology.'

1.3 Solution approach

- **Crop Recommondation** We utilized a dataset containing essential agricultural parameters such as nutrient levels (N, P, K), temperature, humidity, and other environmental factors related to crop cultivation. Based on this data, we implemented and trained three machine learning models: Logistic Regression, Random Forest, and Decision Tree to perform crop prediction.

  To assess the performance of these models, we employed standard evaluation metrics, including:

  - Accuracy
  - Precision
  - Recall
  - F1 Score
  - Confusion Matrix

  These metrics provided a comprehensive understanding of each model's classification capability. Based on the model predictions, the system ultimately recommends the most suitable crop to cultivate under the given soil and climatic conditions.

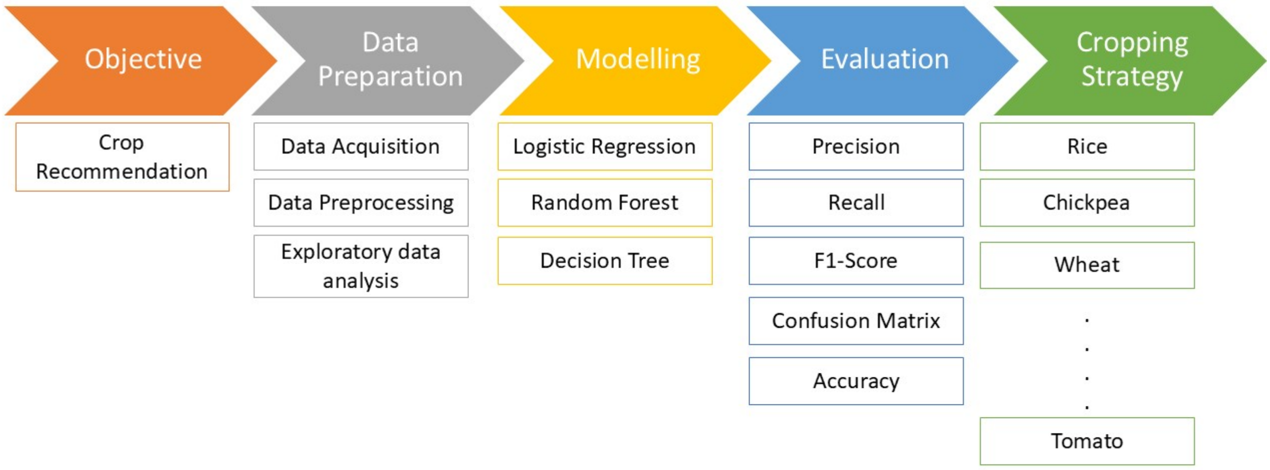

Figure 1.1: Solution Approach for Crop Recommondation

- **Disease Detection**

  In the disease detection module, we focused on identifying plant diseases using image classification techniques applied to five major crops: potato, tomato, onion, rice, and wheat. The input to the system consists of leaf images for each crop, which were used to train and evaluate deep learning models.

  For this task, we developed and compared the performance of two models:

  - A custom Convolutional Neural Network (CNN) designed and trained from scratch.
  - A pre-trained GoogLeNet (Inception v1) model, fine-tuned on the dataset to leverage transfer learning.

  The image data underwent preprocessing steps such as resizing, normalization, and augmentation to enhance the model's ability to generalize. Both models were trained to classify the disease categories specific to each crop.

  To evaluate the performance, we used metrics such as:

  - Accuracy
  - Confusion Matrix

  These metrics provided insight into how well each model could distinguish between different plant diseases. Based on the evaluation, the trained model predicts the specific disease affecting a given leaf image, thereby enabling timely intervention and improved crop health management.

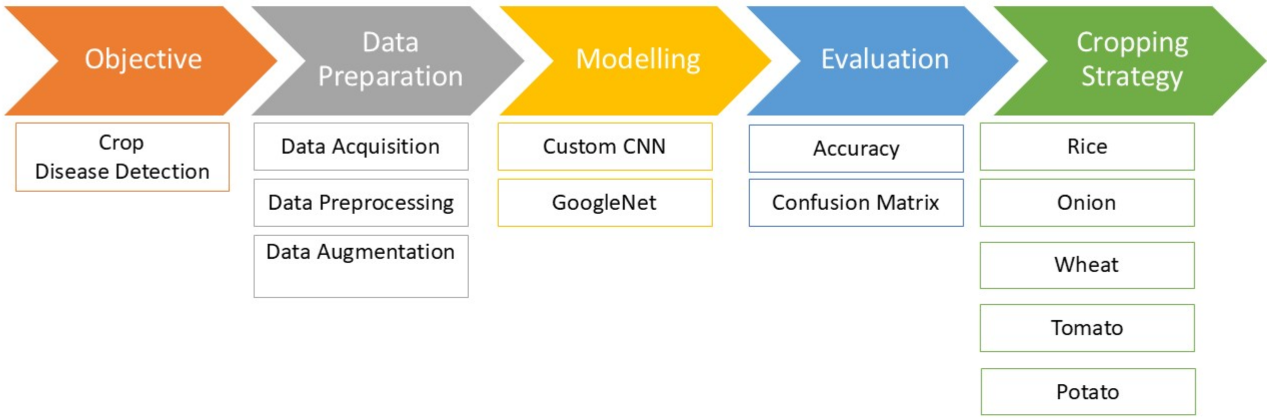

Figure 1.2: Solution Approach for Disease Detection

- **Smart Irrigation** The goal of this project is to develop a smart irrigation recommendation system using historical weather and soil moisture data. The system identifies whether irrigation is needed for specific crops, considering soil moisture, rainfall patterns, temperature, crop type, and geographical location. Machine learning and explainability tools are used to build, train, evaluate, and interpret the model.

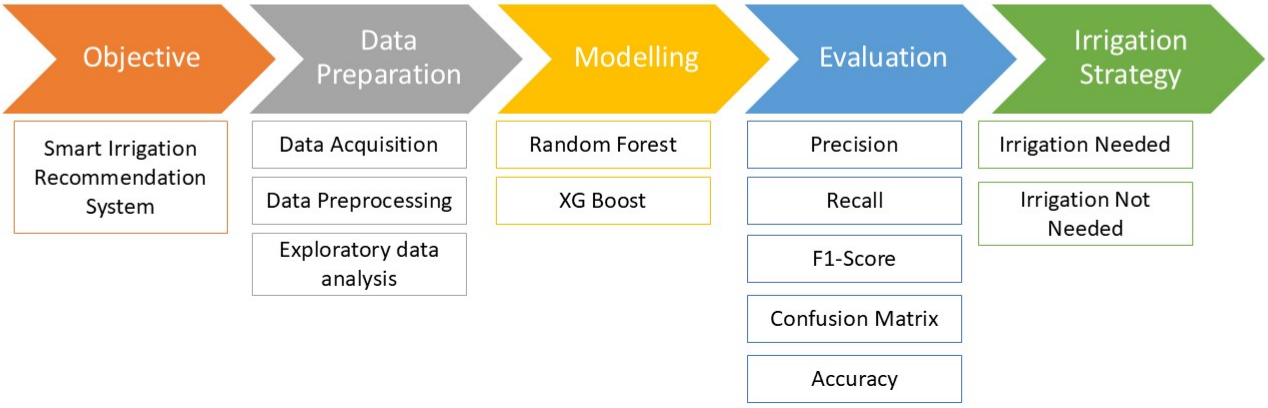

Figure 1.3: Solution Approach for Third objective

Chapter 2

Data Preparation

2.1 Data Acquisition

- **Crop Recommondation** To build and train the machine learning models for the crop recommendation module in AgroAI, we have used the publicly available dataset titled "Smart Crop Prediction", which can be accessed from Figshare[]. This dataset is originally sourced and copyrighted by the Indian Council of Food and Agriculture (ICFA). This is a CSV file that contains several horticulture and agriculture crops.
  Features
  - **N (Nitrogen)**: Represents the nitrogen content in the soil, an essential macronutrient for plant growth.
  - **P (Phosphorus)**: Indicates the phosphorus level, crucial for energy transfer and root development in crops.
  - **K (Potassium)**: Denotes the potassium content, which contributes to disease resistance and water uptake.
  - **Temperature**: The average temperature (in °C) of the environment in which the crop is grown.
  - **Humidity**: The relative humidity (in %) of the air, which affects transpiration and growth conditions.
  - **pH**: The pH level of the soil, indicating whether it is acidic, neutral, or alkaline.
  - **Rainfall**: The amount of rainfall (in mm) received, which is vital for water availability and crop planning.
  - **Label**: The target variable indicating the type of crop (e.g., `rice`, `onion`).
- **Disease Detection** The image data used for training the disease detection model was sourced from the PlantVillage dataset, which is publicly available on Mendeley Data. This dataset comprises a total of 9,821 labeled images representing 18 different plant diseases affecting various crops such as tomato, potato, wheat, rice, and onion. Each image is annotated with its corresponding disease category, making the dataset highly suitable for supervised deep learning tasks. In the PlantVillage image dataset, the features necessary for disease classification—such as color patterns, textures, leaf edges, and lesion shapes—are not manually specified. Instead, these visual features are automatically extracted by the convolutional layers of the deep learning model during the training process. This process enables the model to learn complex and abstract representations of plant diseases directly from raw image pixels, making manual feature

engineering unnecessary. There are 18 different classes that we study which are following :

1. Bacterialblight rice	2. BlackPoint wheat	3. Brownspot rice
4. FusariumFootRot wheat	5. HealthyLeaf wheat	6. Iris yellow virus_onion
7. LeafBlight wheat	8. Leafsmut rice	9. Potato___Early_blight
10. Potato___Late_blight	11. Potato___healthy	
12. Tomato___Early_blight	13. Tomato___Late_blight	14. Tomato___healthy
15. WheatBlast	16. healthy_onion	17. purple blotch onion
18.Stemphylium leaf blight and collectrichum leaf blight onion		

Some of these diseases are shown in the following images.

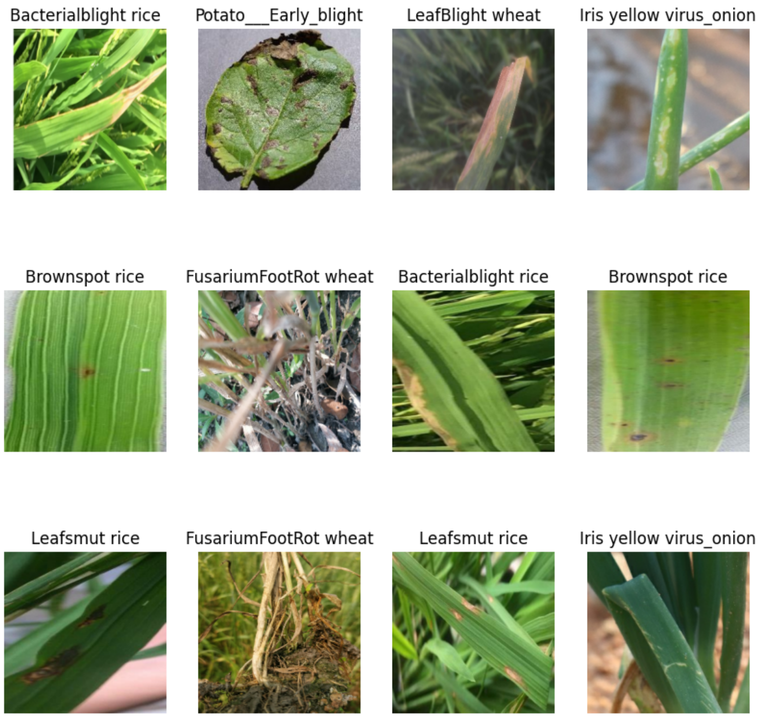

Figure 2.1: Pictorial Illustration of diseases in plants

- **Smart Irrigation**This project utilizes a dataset that focuses on various environmental and agricultural parameters crucial for determining irrigation needs. The dataset encompasses key features related to weather conditions, soil moisture, and crop characteristics, which play an essential role in making irrigation decisions. The data is sourced from a combination of trusted agricultural and meteorological agencies, and it helps in predicting irrigation requirements across different regions in India.

  Features

  - **RainfallValue**:
    * This feature records the amount of rainfall (in millimeters) for each state on a given date. It is crucial for irrigation decision-making because insufficient rainfall leads to dry soil conditions, necessitating additional irrigation to maintain crop health.
    * Source: India Meteorological Department (IMD): Provides daily, monthly, and annual rainfall records for various states in India.

- **Tmax and Tmin(Temprature)**
  * These features capture the maximum and minimum daily temperatures, measured in degrees Celsius. Temperature data is essential for understanding crop growth cycles, evaporation rates, and the overall climatic conditions that influence soil moisture content. High temperatures can lead to increased evaporation, while low temperatures can slow down evaporation.
  * Source: India Meteorological Department (IMD): Offers temperature data across India, including both Tmax (maximum) and Tmin (minimum) daily readings.
- **Soil Moisture**
  * Soil moisture represents the percentage of water content in the soil, which is an indicator of how much moisture the soil retains at any given time. Low soil moisture typically indicates the need for irrigation, especially when rainfall is insufficient.
  * Source:SMAP L4 Daily Product at 9Km Resolution: The SMAP (Soil Moisture Active Passive) mission provides high-resolution soil moisture data from satellite observations, which is used to estimate the moisture content in the soil. The L4 daily product is a gridded dataset with a 9km resolution, providing accurate soil moisture estimates at a global scale.
- **Crop Type (Wheat, Rice, Potato, Tomato)**
  * The type of crop grown in a state significantly influences its water requirements. Different crops have distinct irrigation thresholds, and this feature allows the model to apply crop-specific irrigation rules and thresholds.
  * ICAR (Indian Council of Agricultural Research): Offers extensive data on crop types and cropping patterns.

This dataset, comprising weather, soil, and crop-specific data, is vital for determining irrigation needs and creating a robust predictive model. The data from IMD (for rainfall and temperature) and NASA SMAP L4 (for soil moisture) allows for a comprehensive view of the environmental and agricultural conditions across India. By processing the daily NetCDF files for rainfall, temperature, and soil moisture data from 2014 to 2024, the project ensures accurate, temporal, and spatial alignment of all features, facilitating the prediction of irrigation requirements based

2.2 Data Processing

- **Crop Prediction** The dataset obtained from Figshare was pre-processed and well-structured. It did not contain any missing or null values, eliminating the need for data cleaning at this stage.

- **Disease Detection** In this part, the entire dataset consists of plant leaf images. Through the preprocessing steps, the images were standardized and augmented, resulting in uniformly sized, normalized, and diversified data. These outcomes significantly enhance the quality of the input data and play a crucial role in improving the performance and generalization of the deep learning model during training.

```
resize_and_rescale = tf.keras.Sequential([
    layers.Resizing(256, 256),
```

```
    layers.Rescaling(1./255)
])

data_augmentation = tf.keras.Sequential([
    layers.RandomFlip("horizontal_and_vertical"),
    layers.RandomRotation(0.2)
])
```

  - Ensures that all input images are the same size (256×256), which is required for CNNs.
  - Rescaling the pixel values helps the model train faster and more accurately by standardizing the input range.
  - Adds variation to the training images, so the model learns to generalize better and doesn't overfit.
  - Helps simulate different orientations and perspectives of the same plant leaves.

- **Smart Irrigation**
  - **Missing Value Treatment**:
    * Several important columns had missing values, such as rainfall, temperature (Tmax and Tmin), and soil moisture.
    * Missing values were filled using the monthly average per state, capturing regional and seasonal patterns.
    * For any values still missing after this, forward fill and backward fill techniques were applied to ensure completeness
  - **Datetime Conversion and Feature Extraction**:
    * The TimeStamp column was converted to a proper datetime format.
    * A new feature, Month, was extracted from the timestamp to support seasonal analysis.
  - **Data Sorting**:
    * The data was sorted by StateName and TimeStamp to maintain chronological integrity, which is crucial for time-based operations.
  - **Rolling Feature Engineering**:
    * A new feature Rainfalllast4months was computed to represent the 4-months cumulative rainfall for each state.
    * This rolling metric reflects short-term water availability and supports dynamic irrigation decisions.
  - **Rule-Based Labeling**:
    * A binary target variable, IrrigationNeeded, was created based on crop-specific thresholds.
    * These thresholds considered minimum required soil moisture and 4-months rainfall totals for crops such as wheat, rice, potato, and tomato.
  - **Data Expansion Across Crops**:
    * The dataset was duplicated across crop types, applying unique threshold rules to each.

    * This allowed the same environmental conditions to be evaluated from the perspective of different crops.
  - **Categorical Encoding**:
    * Categorical variables (StateName, Crop) were converted into numerical codes using label encoding.
    * This step ensured the data could be used with machine learning models that require numeric input.
  - **Final Feature Set Selection**:
    * The final dataset included key features like rainfall, temperature, soil moisture, encoded state/crop, month, and the newly engineered rolling rainfall metric.

2.3 Exploratory Data Analysis

2.3.1 Crop Recommondation

Dataset Description The dataset consists of 2500 rows and 8 columns, capturing various environmental and soil-related features alongside the corresponding crop label. Specifically, there are three integer-type columns (N, P, K), four floating-point columns (temperature, humidity, ph, rainfall), and one categorical column (label), which represents the crop type.

	N	P	K	temperature	humidity	ph	rainfall	label
0	90	42	43	20.879744	82.002744	6.502985	202.935536	rice
1	85	58	41	21.770462	80.319644	7.038096	226.655537	rice
2	60	55	44	23.004459	82.320763	7.840207	263.964248	rice
3	74	35	40	26.491096	80.158363	6.980401	242.864034	rice
4	78	42	42	20.130175	81.604873	7.628473	262.717340	rice
...	...	...	...	...	...	...	...	...
2495	118	23	77	19.470000	72.050000	6.370000	723.680000	onion
2496	117	21	47	21.210000	67.490000	6.480000	705.770000	onion
2497	121	25	62	20.390000	68.300000	5.860000	708.650000	onion
2498	143	25	69	24.330000	69.340000	6.350000	706.450000	onion
2499	142	20	75	24.330000	67.540000	6.210000	687.880000	onion

2500 rows × 8 columns

Figure 2.2: DATASET

2.3.2 Dataset Information

The dataset contains a total of 2500 entries and 8 columns. The details of each column, including data types and non-null counts, are as follows:

#	Column	Non-Null Count	Dtype
0	N	2500	int64
1	P	2500	int64
2	K	2500	int64
3	temperature	2500	float64
4	humidity	2500	float64
5	ph	2500	float64
6	rainfall	2500	float64
7	label	2500	object

Table 2.1: Column-wise summary of the dataset

All columns are complete with no missing values. There are three integer columns, four float columns, and one categorical (object) column.

Unique Crop Labels

The dataset contains the following unique crop labels:

`rice, maize, chickpea, kidneybeans, pigeonpeas, mothbeans, mungbean, blackgram, lentil, pomegranate, banana, mango, grapes, watermelon, muskmelon, apple, orange, papaya, coconut, cotton, jute, coffee, tomato, potato, onion`

2.4 Insights from Scatterplot Matrices of Crop Categories

Two separate scatterplot matrices were generated to analyze the relationships between key environmental and nutrient features for agriculture and horticulture crops. These plots help visualize the feature space each crop occupies and provide insights into their growing conditions.

Agriculture Crops

The scatterplot matrix for agriculture crops (e.g., rice, maize, pulses, cotton, jute) reveals the following insights:

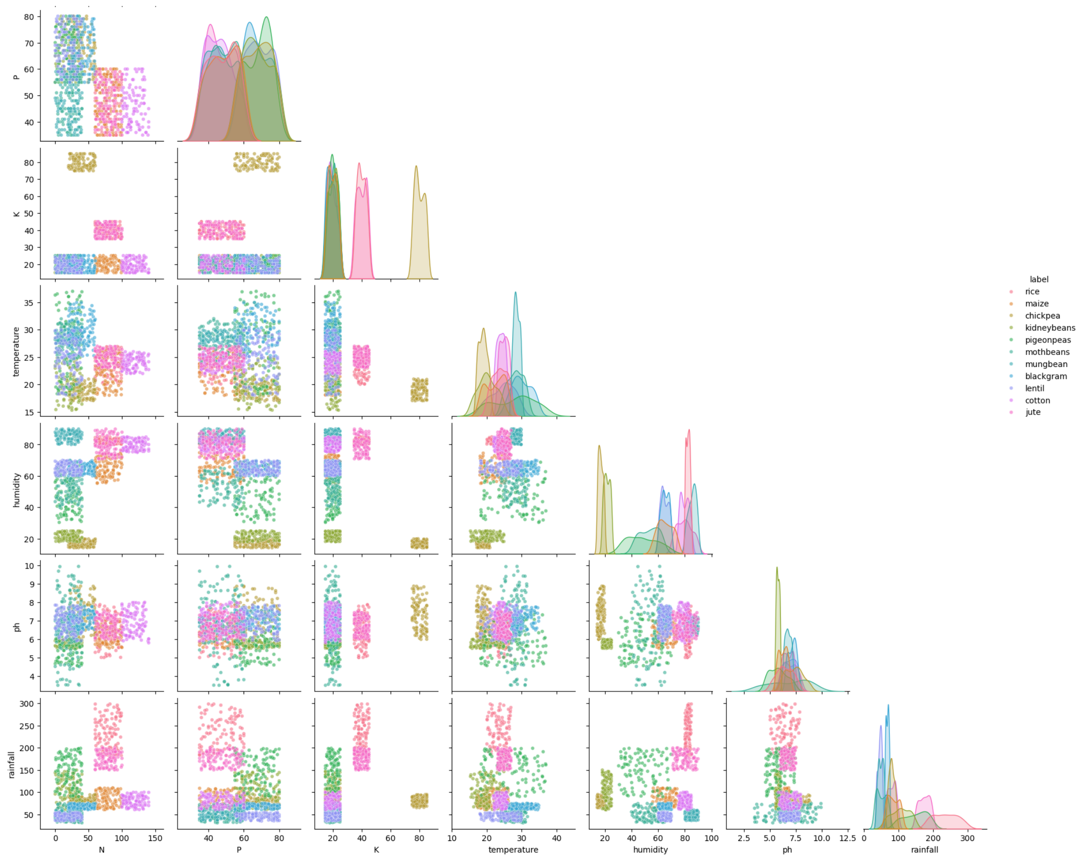

Figure 2.3: Scatterplot matrix for agriculture crops

- **Clustered Groups by Crop:** Crops such as *cotton*, *jute*, and *maize* exhibit well-separated clusters across multiple features, indicating clearly distinguishable growing conditions.
- **Strong Grouping in Temperature and Humidity:** Temperature and humidity are significant differentiators:
  - *Cotton* prefers high temperatures but lower humidity.
  - Pulses such as *lentil*, *blackgram*, and *mungbean* cluster in moderate temperature and humidity ranges.
- **Nutrient Requirements:**
  - *Jute* requires high nitrogen levels.
  - *Chickpea* and *lentil* are associated with lower phosphorus and potassium needs.
- **Overlap Among Pulses:** Pulses like *mungbean*, *blackgram*, and *kidneybeans* exhibit overlapping regions in terms of pH and rainfall, suggesting similar growing environments.

Horticulture Crops

The scatterplot matrix for horticulture crops (e.g., fruits, vegetables, and plantation crops) reveals greater diversity in feature requirements:

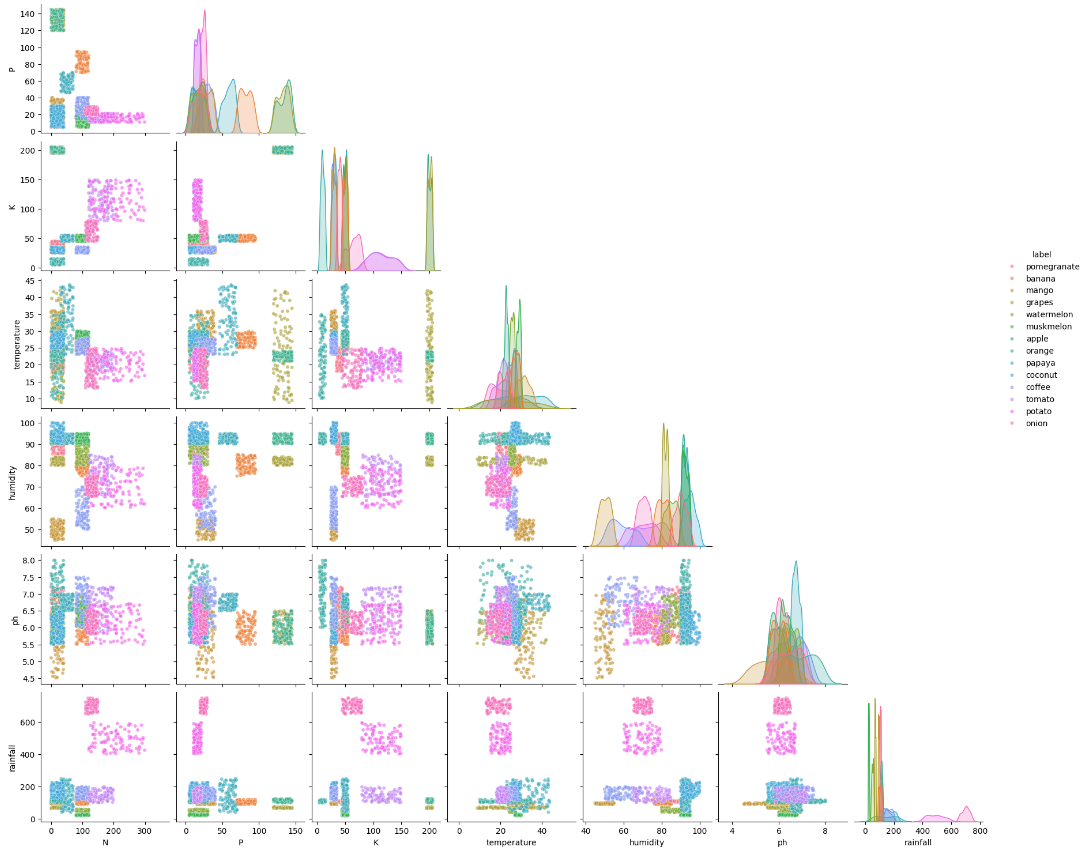

Figure 2.4: Scatterplot matrix for Horticulture crops

- **Rainfall Variation:** Rainfall is a major distinguishing factor:
  - *Coffee* and *coconut* demand high rainfall.
  - *Tomato*, *onion*, and *potato* thrive in relatively low rainfall zones.
- **Distinct Clusters:**
  - *Pomegranate*, *grapes*, and *muskmelon* form tight, isolated clusters.
  - *Tomato*, *potato*, and *onion* have some overlap but are separable using temperature and nutrients.
- **Temperature Preferences:**
  - *Papaya*, *banana*, and *mango* prefer higher temperatures.
  - *Apple* and *grapes* thrive in cooler temperature zones.
- **pH Range:** Most horticulture crops grow best in a pH range of 5.5–7.0. However, crops like *grapes* and *coconut* show preference for specific pH values, slightly outside this common range.

2.4.1 Comparative Analysis of Agricultural and Horticultural Crops Based on Environmental and Nutrient Parameters

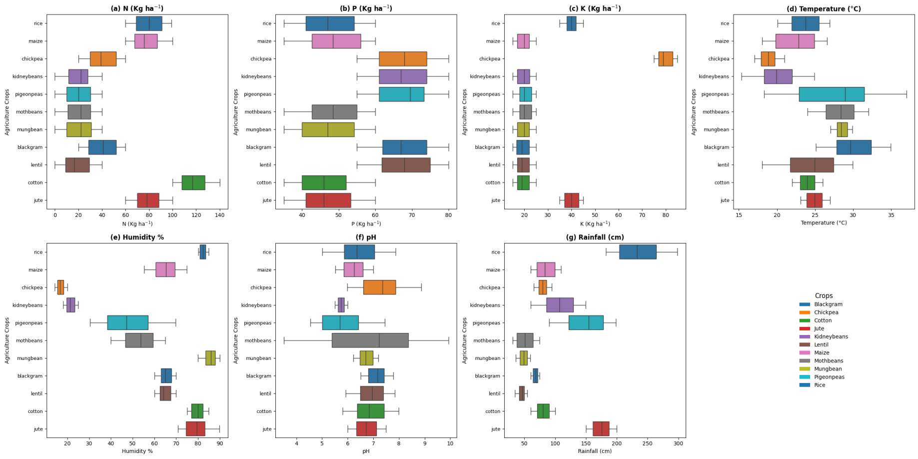

Figure 2.5: BOX-PLOTS-AGRICULTURE

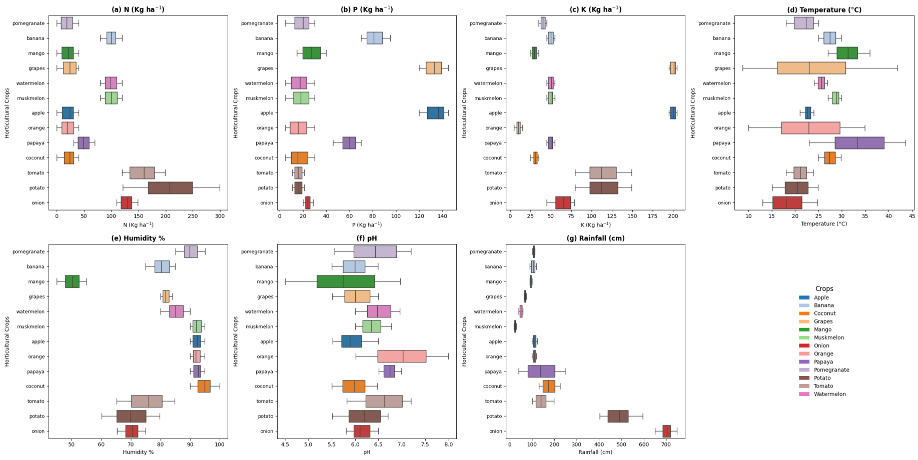

Figure 2.6: BOX-PLOTS-HORTICULTURE

Overview

Figures 2.5 and 2.6 present boxplots visualizing the distribution of key parameters—Nitrogen (N), Phosphorus (P), Potassium (K), Temperature, Humidity, pH, and Rainfall—for a range of agricultural and horticultural crops. These plots offer insights into the distinct agro-environmental needs and tolerances of the two crop categories.

1. Nutrient Requirements (N, P, K)

Agricultural Crops: - Most agricultural crops exhibit moderate nitrogen requirements in the range of 20–100 Kg/ha, with *jute* having the highest nitrogen demand. - Phosphorus needs remain relatively consistent across crops (40–80 Kg/ha), with slight variations for crops like *cotton* and *lentil*. - Potassium requirements are generally low, with a few outliers such as *jute* needing above 80 Kg/ha.
Horticultural Crops: - Horticultural crops show a broader range of nutrient requirements. -

Potato and *onion* demand the highest nitrogen (up to 250 Kg/ha), phosphorus, and potassium. - *Fruit crops* like *banana*, *mango*, and *pomegranate* require lower NPK compared to root vegetables.

2. Temperature (°C)

Agricultural Crops: - Most crops fall within a narrow temperature window of 20–30°C. - *Pigeonpeas* and *mungbeans* tolerate slightly higher temperatures (35°C).
Horticultural Crops: - A more diverse temperature profile is observed. - *Onion* and *potato* perform well in cooler climates (15–25°C). - *Papaya*, *mango*, and *pomegranate* require higher temperatures, ranging from 25–40°C.

3. Humidity (%)

Agricultural Crops: - Humidity preferences vary widely. - *Maize* and *chickpea* prefer dry climates (20–30%), while *rice* and *jute* favor high humidity environments (70–90%).
Horticultural Crops: - Most fruit crops prefer higher humidity (70–90%). - *Potato* and *onion* thrive in relatively lower humidity (60–70%).

4. Soil pH

Agricultural Crops: - The pH range is narrow (5.5–7.5), with most crops preferring neutral to slightly acidic soil.
Horticultural Crops: - A wider pH variation is visible. - *Banana*, *mango*, and *papaya* tolerate slightly more acidic conditions (pH 5.0–6.0), while *grapes* and *apple* prefer near-neutral soils.

5. Rainfall (cm)

Agricultural Crops: - Crops like *rice* require very high rainfall (250–300 cm). - Others like *chickpea* and *lentil* are adapted to lower rainfall levels (50–100 cm).
Horticultural Crops: - *Onion* and *potato* require rainfall exceeding 600 cm. - Fruit crops like *papaya*, *mango*, and *banana* prefer 100–300 cm.

Conclusion

This comparison reveals significant differences between agricultural and horticultural crops. Agricultural crops show relatively uniform environmental requirements, making them more adaptable to standardized farming practices. In contrast, horticultural crops exhibit greater variability in nutrient, temperature, and rainfall needs, requiring more tailored agronomic planning for optimal yield. These distinctions are crucial for crop selection, land suitability analysis, and precision farming strategies.

2.4.2 Correlation Analysis

The heatmap in Figure **??** presents the Pearson correlation coefficients among key environmental and soil-related variables: **N**, **P**, **K** (macronutrients), **temperature**, **humidity**, **pH**, and **rainfall**. The correlation coefficient values range from -1 to 1, where:

- $+1$ indicates a perfect positive linear relationship,
- -1 indicates a perfect negative linear relationship,

- 0 indicates no linear correlation.

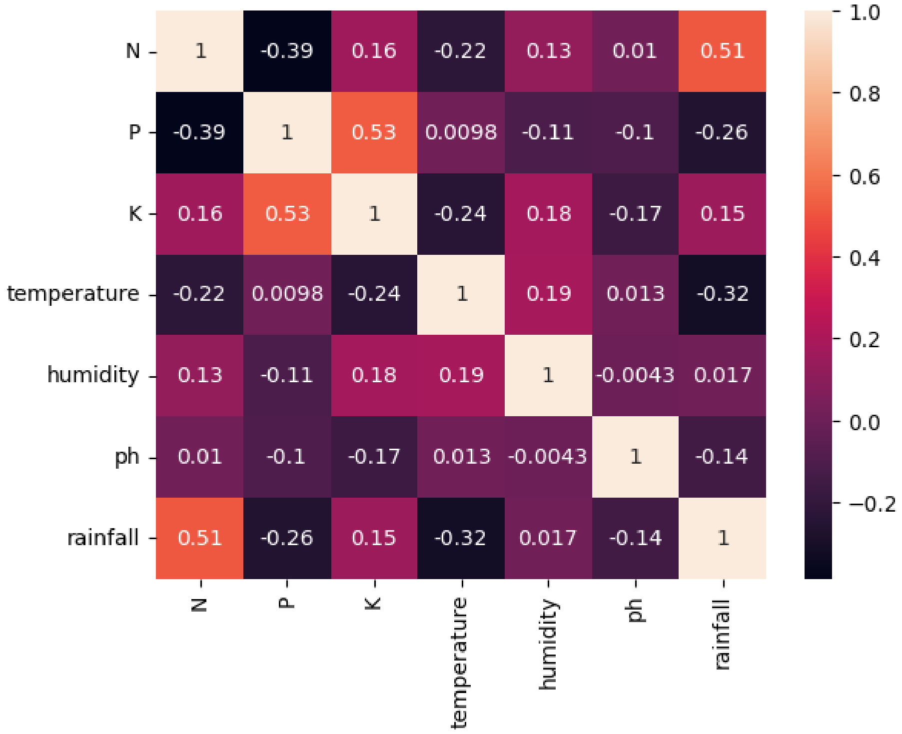

Figure 2.7: BOX-PLOTS-AGRICULTURE

Key Observations:

1. **N and Rainfall (**0.51**)**: Moderately positive correlation, suggesting that higher nitrogen levels are associated with higher rainfall. This may be due to leaching or enhanced nitrogen uptake during rainy seasons.

2. **P and K (**0.53**)**: Strong positive correlation, indicating these two nutrients often co-occur or are similarly influenced by soil properties or fertilization practices.

3. **Temperature shows weak negative correlation with nutrients**: With K (-0.24), N (-0.22), and Rainfall (-0.32), temperature has a slightly inverse relationship, possibly implying reduced nutrient availability or uptake at higher temperatures.

4. **Humidity is weakly correlated with all variables**: The values are close to zero (maximum 0.19 with temperature and K), indicating minimal linear relationships.

5. **pH and other variables**: Correlations are negligible (e.g., -0.17 with K, -0.14 with rainfall), suggesting that pH is relatively independent in this dataset.

6. **P and N (**-0.39**)**: A moderate negative correlation, potentially indicating an inverse relationship or trade-off between phosphorus and nitrogen levels, possibly due to fertilization regimes or soil chemistry.

Conclusion: While most variables exhibit weak correlations, a few relationships stand out—notably **P–K** and **N–Rainfall**—which merit further exploration. These findings can inform better soil nutrient management and climate-aware agricultural practices.

2.4.3 Smart Irrigation

The analysis reveals significant regional and seasonal variations in rainfall across India. Northeastern and coastal states receive the highest rainfall, while interior regions like Karnataka, Rajasthan, and Punjab experience much lower levels. A clear monsoon cycle is evident, with peak rainfall during June–September and minimal precipitation in winter months, highlighting challenges for water and agricultural planning. Temperature patterns show strong seasonal fluctuations, with Tmax peaking in summer and dropping in winter. Correlation analysis indicates that rainfall is closely linked to soil moisture but largely independent of Tmax, while Tmin shows strong alignment with Tmax and weaker links with other variables.

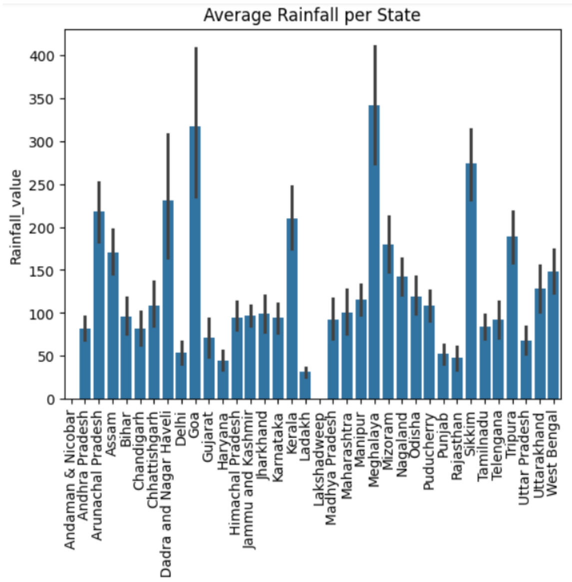

Figure 2.8: Average rainfall per state VS rainfallvalue

This figure illustrates the average rainfall across various Indian states and union territories. Each blue bar represents a state's average rainfall, while the black lines indicate the error margins. Notably, Meghalaya records the highest average rainfall (approximately 340–350 mm), followed by Goa (around 315–320 mm) and Uttarakhand (270–280 mm). On the lower end, Karnataka shows the least rainfall (around 30–40 mm), with Rajasthan and Punjab also receiving relatively low amounts (about 50–60 mm). Overall, the chart highlights substantial regional variation, with northeastern and coastal states experiencing significantly more rainfall than interior areas.

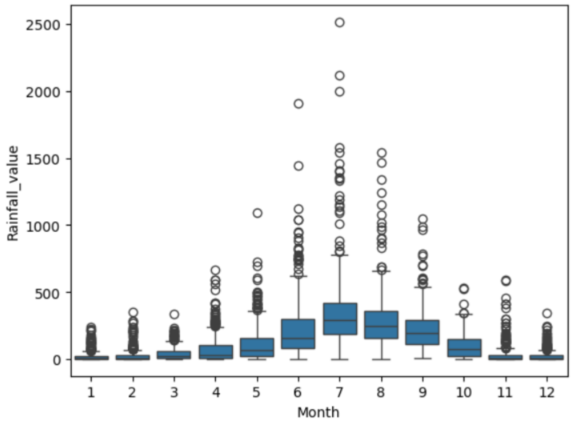

Figure 2.9: Month VS rainfallvalue Boxplot

This boxplot illustrates monthly rainfall patterns, highlighting a distinct monsoon cycle. Rainfall is minimal from January to March, rises from April, and peaks in July with the highest median and variability. The monsoon period (June–September) shows both the most rainfall and frequent extreme outliers. Rainfall then declines through November, returning to low levels by December. The pattern reflects a region with pronounced wet and dry seasons, posing challenges for water management and agriculture.

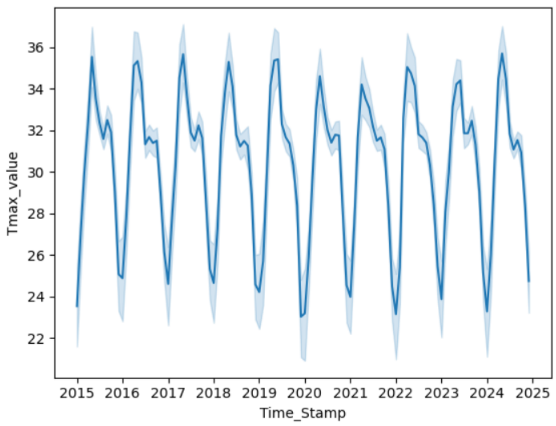

Figure 2.10: Time Stame VS Tmax value

This figure shows a time series plot of maximum temperature (Tmax) values from 2015 to 2025. The dark blue line represents the actual temperature measurements, while the light blue shaded area likely represents the confidence interval around these measurements. The data displays a clear seasonal pattern with temperatures cycling regularly throughout each year. The peaks (around 35-36°C) appear to occur annually, likely during summer months, while the valleys (around 23-25°C) represent cooler periods, probably winter months. The data also shows some secondary peaks and dips within each annual cycle, possibly representing shorter-term seasonal variations or weather events within each year.

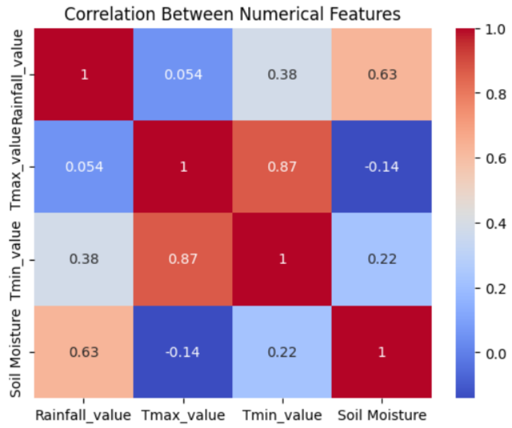

Figure 2.11: Time Stame VS Tmax value

The correlation matrix shows that rainfall and maximum temperature (Tmax) are largely uncorrelated (0.054), while minimum temperature (Tmin) and Tmax have a strong positive correlation (0.87). Soil moisture is moderately correlated with rainfall (0.63) and slightly negatively correlated with Tmax (-0.14). Weak positive correlations are observed between rainfall and Tmin (0.38), and between soil moisture and Tmin (0.22).

Chapter 3

Models

3.1 Crop Recommendation

3.1.1 Model Evaluation

To evaluate the performance of the crop recommendation system, three classification models—Logistic Regression, Decision Tree, and Random Forest—were trained and tested. Among them, the **Random Forest classifier** demonstrated superior performance with an overall **accuracy of 99.60%**, indicating excellent classification capability across all crop classes. The **Logistic Regression model** also performed well, achieving a respectable **accuracy of 96.4%**. In contrast, the **Decision Tree model**, while still effective, yielded a relatively lower **accuracy of 89.8%**. These results clearly highlight the Random Forest model as the most reliable and robust choice for crop prediction in this system, offering high accuracy and consistent performance across different crop categories.

Table 3.1: Performance Metrics of Classification Models for Crop Recommendation

Model	Accuracy (%)	Precision (Macro)	Recall (Macro)	F1 Score (Macro)
Logistic Regression	96.4	0.9613	0.9639	0.9608
Decision Tree	89.8	0.8764	0.9024	0.8705
Random Forest	99.6	0.9963	0.9963	0.9963

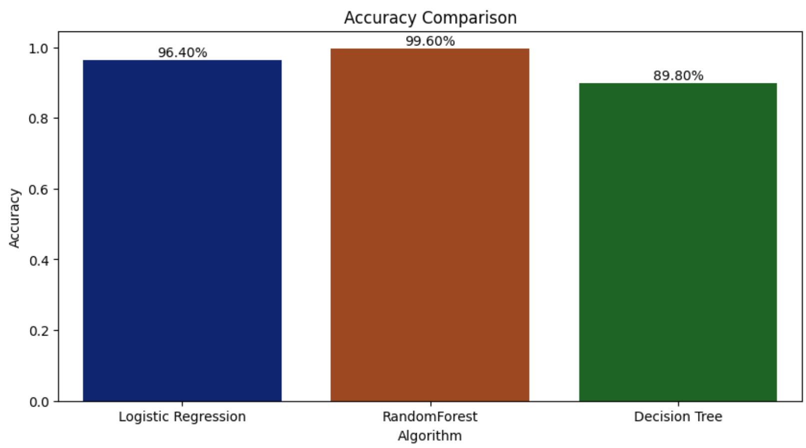

Figure 3.1: Accuracy Comparison

3.1.2 Analysis of confusion matrix to visualize the recommendationresponse of models

In this study, three machine learning models—**Logistic Regression**, **Decision Tree**, and **Random Forest**—were evaluated for their effectiveness in a crop recommendation system using multiple performance metrics including *Accuracy*, *Precision*, *Recall*, *F1 Score*, and visual analysis through confusion matrices. The **Random Forest** model demonstrated the best performance, achieving an accuracy of **99.6%**, along with a macro precision, recall, and F1 score of **0.9963** each. Its confusion matrix showed near-perfect predictions with almost all values aligned along the diagonal, indicating minimal misclassifications and excellent generalization. **Logistic Regression** also performed strongly, with an accuracy of **96.4%**, macro precision of **0.9613**, recall of **0.9639**, and F1 score.

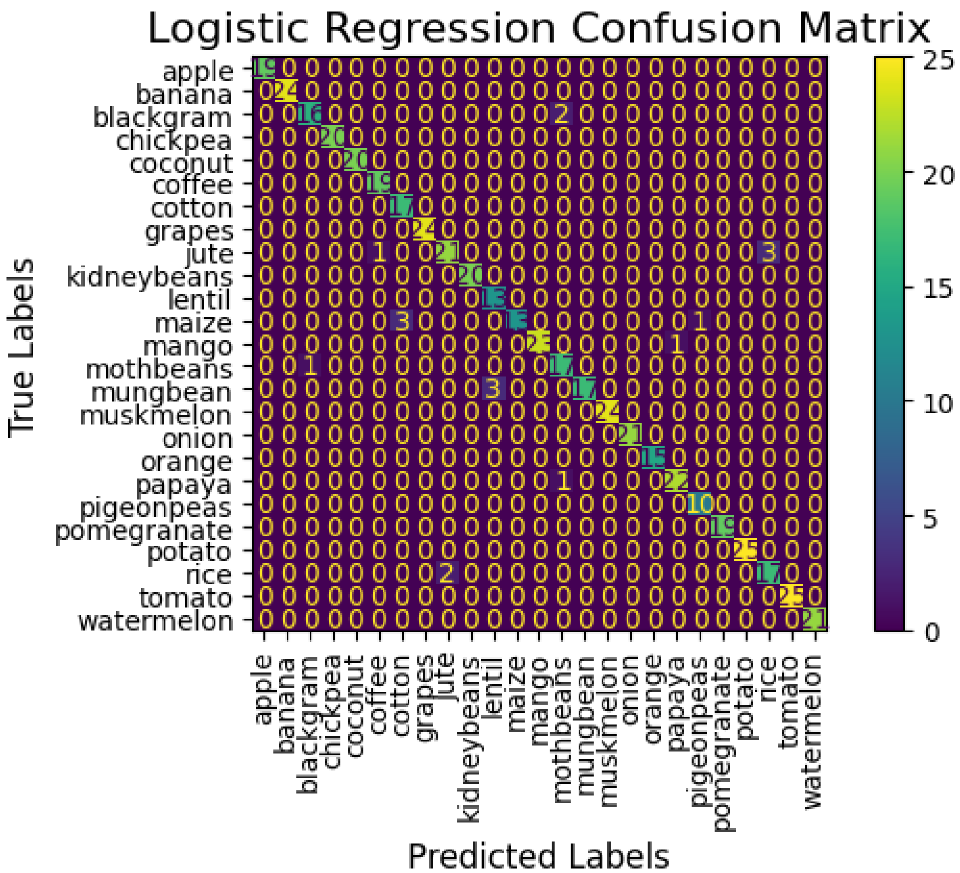

Figure 3.2: Logistic Regression Confusion Matrix

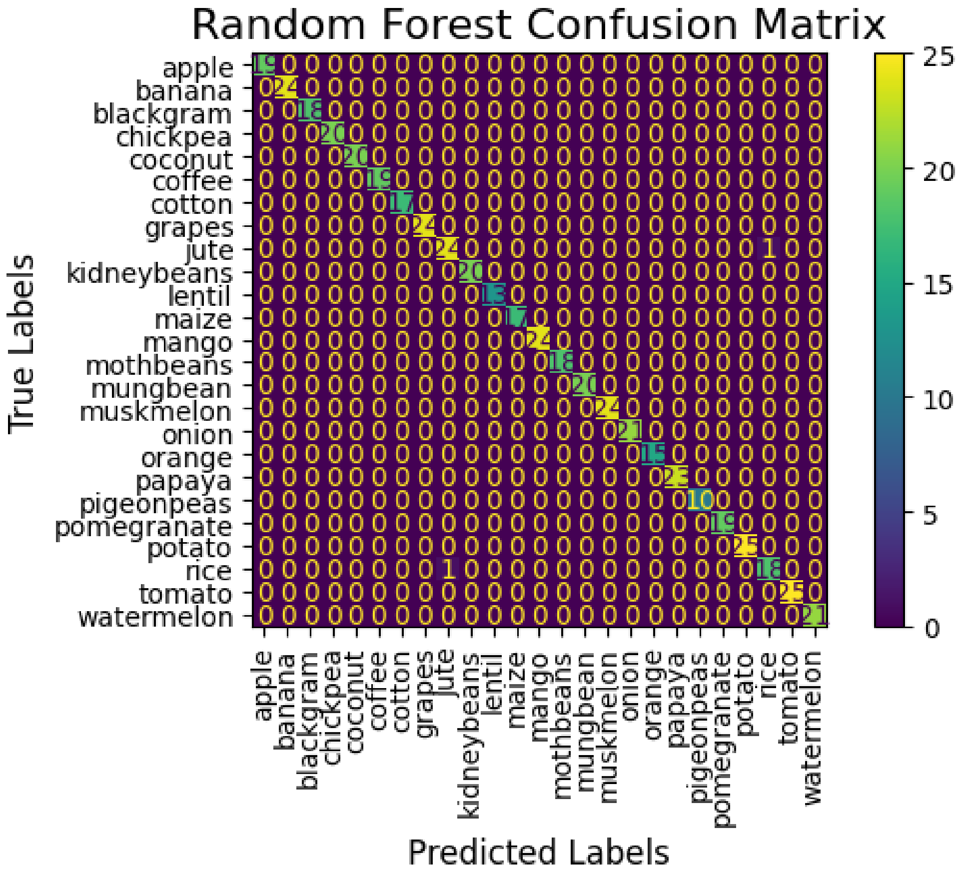

Figure 3.3: Random Forest Confusion Matrix

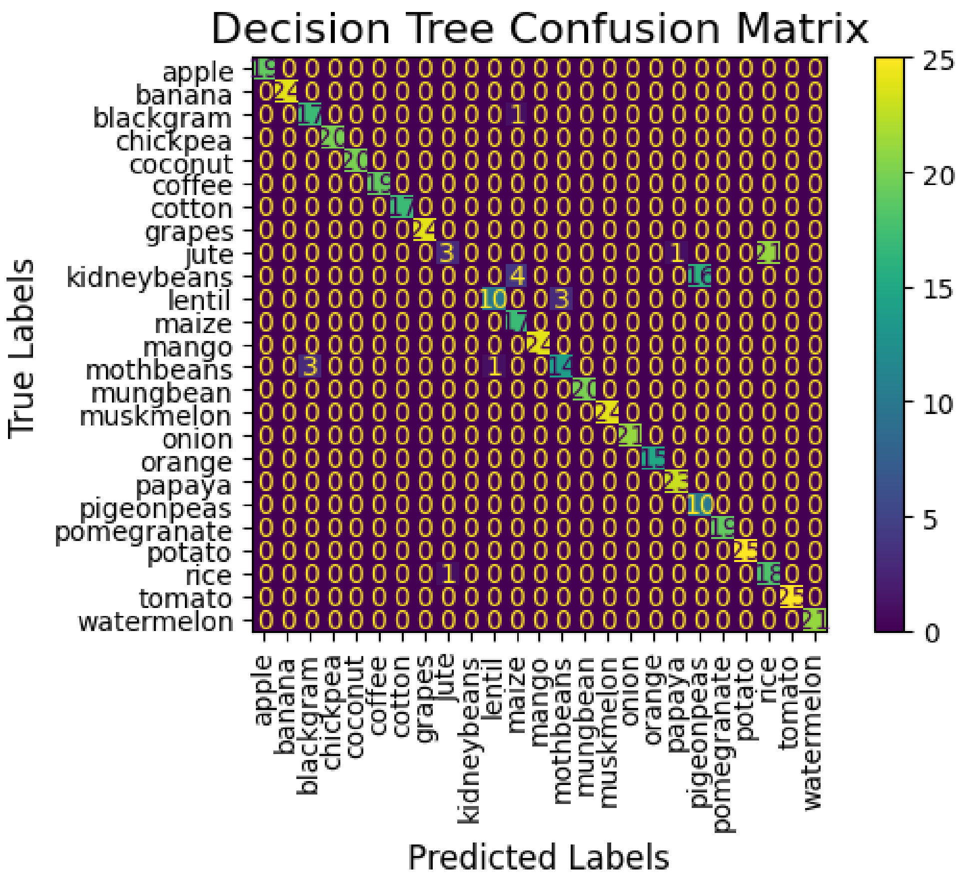

Figure 3.4: Decision Tree Confusion Matrix

3.2 Disease Detection

This report details the development and performance of two deep learning models for plant disease classification: a custom CNN architecture and a fine-tuned GoogleNet model. The system achieves 92.8% accuracy in 18 disease classes that contains around 9821 images (5 different categories of plants) from the Plant Village data set namely:

3.2.1 CNN Architecture

Layer Breakdown

Table 3.2: Custom CNN Architecture Breakdown

Block	**Layers**	**Parameters / Description**
Input	Resize / Rescale	256×256×3 image input, normalized to [0, 1]
Block 1	Conv2D(32) + BatchNorm	3×3 kernel, 32 filters, basic feature extraction
	MaxPool + Dropout	2×2 pooling, 25% dropout to reduce overfitting
Block 2	Conv2D(64) + BatchNorm	3×3 kernel, 64 filters, deeper features
	MaxPool + Dropout	2×2 pooling, 35% dropout
Block 3	Conv2D(128) + BatchNorm	3×3 kernel, 128 filters, high-level features
	GAP + Dropout	Global Average Pooling + 50% dropout
Classifier	Dense(1024→18) + Softmax	Fully connected layer, softmax for class probabilities

3.2.2 Training Configuration

Optimization

To train the custom CNN effectively, we utilized the Adam optimizer with a learning rate of 0.0005, which is known for adaptive learning rate adjustments and fast convergence.

To further improve training efficiency and prevent overfitting, the following callbacks were integrated:

- EarlyStopping with a patience of 8 epochs, which stops training when the validation accuracy ceases to improve.
- ReduceLROnPlateau with a reduction factor of 0.2, which lowers the learning rate when the model performance plateaus.

The training was conducted using a batch size of 32 to balance learning stability and computational efficiency.

Performance Metrics

The table below highlights the training progress at selected epochs. It shows steady improvement in both training and validation accuracy, alongside a consistent decrease in validation loss, indicating that the model learned effectively without overfitting.

Table 3.3: Training Progress Over Epochs

Epoch	Training Accuracy	Validation Accuracy	Validation Loss
1	0.35	0.16	2.75
10	0.72	0.60	1.28
20	0.81	0.70	1.12
30	0.86	0.74	0.96

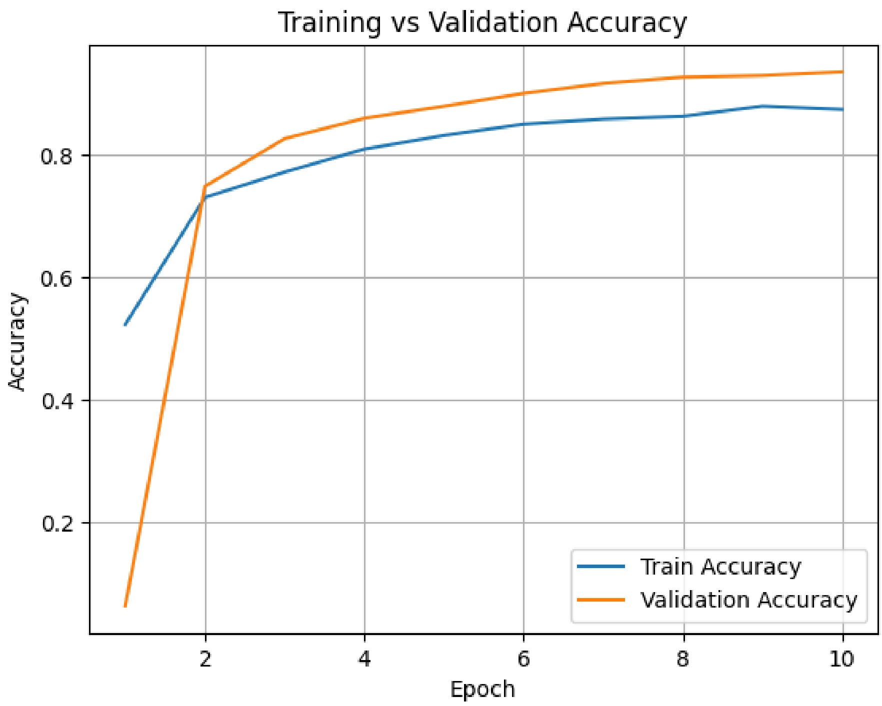

Figure 3.6: Training vs Validation Accuracy

3.2.3 Model Comparison

To evaluate the effectiveness of different architectures in plant disease classification, we compared a Custom Convolutional Neural Network (CNN) with a pre-trained GoogleNet model. The table below summarizes the key performance and architectural differences between the two models:

Table 3.4: Comparison of Custom CNN and GoogleNet Architectures

Metric	Custom CNN	GoogleNet
Test Accuracy	85.2%	92.8%
Training Epochs	30	16
Model Parameters	4.2 Million	6.8 Million
Inference Speed	Faster	Slower

Interpretation

- The GoogleNet model outperformed the custom CNN in terms of test accuracy, achieving 92.8% compared to 85.2%.
- GoogleNet converged faster, requiring only 16 epochs compared to 30 epochs for the custom CNN.
- However, the Custom CNN has fewer parameters and faster inference speed, making it more suitable for resource-constrained deployments such as mobile or edge devices.

3.2.4 Confusion Matrix Analysis

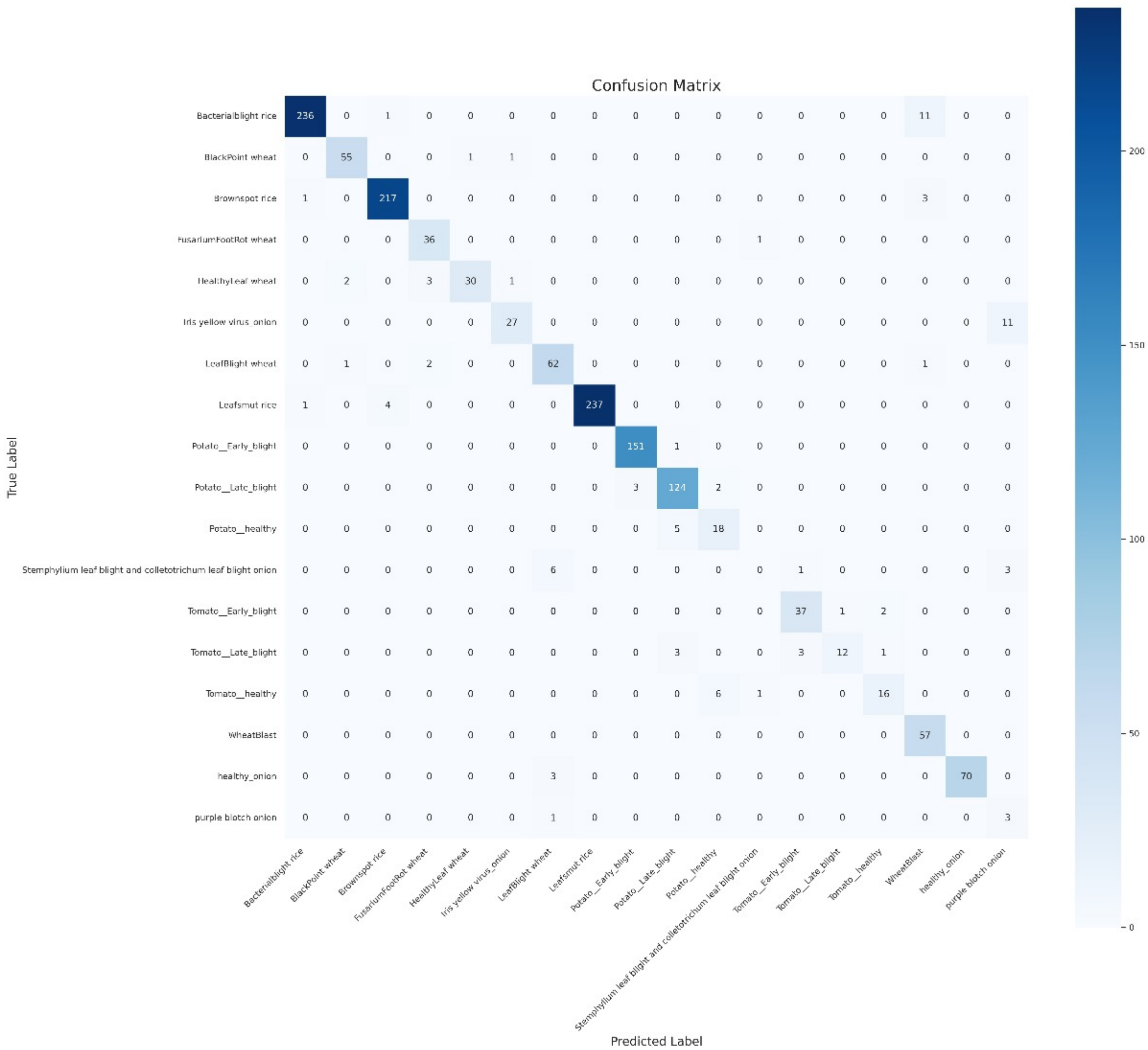

Figure 3.7: Confusion Matrix

Key Observations

1. **High Accuracy Classes:**

- Leaf smut rice: 237 correct predictions, almost no misclassifications.
- Potato-Late-blight: 124 correct predictions, with very few errors.
- Wheat rust: 57 correct predictions and minimal confusion.

2. **Classes With Some Confusion:**
   - Potato-healthy: 38 correct predictions, but some samples confused with Potato-Late-blight and Potato-Early-blight.
   - Tomato-Early-blight and Tomato-Late-blight: Some confusion between each other due to visual similarity.

3. **Minor Misclassifications:**
   - Stemphylium and Colletotrichum onion: Some overlap with Purple blotch onion.
   - Healthy onion: Mostly predicted correctly (70), with very low confusion.

3.3 Smart Irrigation

3.3.1 Machine Learning Models

For this project, two machine learning models were selected: Random Forest Classifier and XGBoost Classifier, both known for their strong performance in classification tasks. The Random Forest Classifier was chosen due to its capability to handle both categorical and numerical data effectively, along with its robustness against overfitting through the use of ensemble learning. The model was trained on the balanced dataset created using SMOTE, with key parameters including n-estimators=100, which defines the number of decision trees in the forest, and random-state=42 to ensure reproducibility of results. The second model, XGBoost Classifier, is a high-performance gradient boosting algorithm widely used in structured data problems. It was configured with eval-metric="logloss" to suit binary classification objectives, while other hyperparameters were kept at their default settings for the initial experimentation phase. XGBoost offers several advantages, including high accuracy, built-in mechanisms to prevent overfitting through regularization, and the ability to handle missing values natively, making it a strong candidate for irrigation prediction tasks.

3.3.2 Model Evaluation

Both models were evaluated on the test set using the following metrics:

- Classification Report: Precision, Recall, F1-score
- Confusion Matrix: To observe the true positive and false positive distribution.
- F1 Score vs Threshold Curve: Used to analyze how the model's performance varies with different probability thresholds.

3.3.3 Threshold Optimization

- By default, classifiers use a threshold of 0.5 to assign class labels.
- However, since this is a domain-sensitive problem (false negatives could lead to crop stress), custom thresholds were selected:
  - Random Forest: 0.43 (optimal F1-score)
  - XGBoost: 0.98 (model showed high confidence in predictions)

3.3.4 Feature Importance

After training the Random Forest model, the importance of each feature was extracted using the .feature-importances- attribute. This provides insight into which features contribute the most to the model's predictions. A horizontal bar chart was used to visualize the feature importance, with features like Soil Moisture, Rainfall, and Crop Code emerging as the most influential. This helps understand what factors are driving the model's decision-making process.

Random Forest Feature Importances: Rainfall over the last 4 months is the most influential feature (46.7%), followed by Rainfall value (19.7%) and Month (16.7%), emphasizing the impact of both cumulative and seasonal rainfall. Soil Moisture (5.9%), State-Code (3.3%), and temperature variables (Tmax: 2.9%, Tmin: 2.7%) have moderate influence, while Crop-Code contributes the least (2.2%). Overall, rainfall-related features dominate the model's predictions.

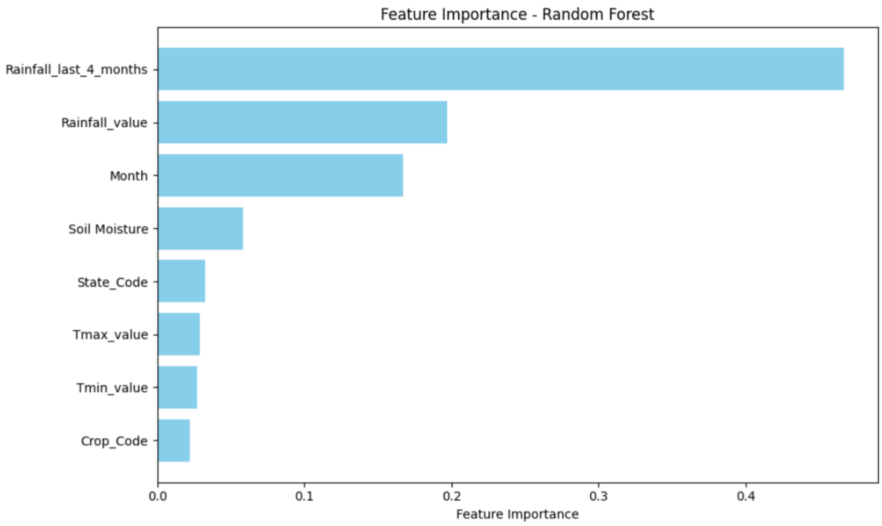

Figure 3.8: Feature importance -Random Forest

XGBoost Feature Importances: The XGBoost model heavily prioritized Rainfall over the last 4 months, assigning it 76.8% of the total importance. Other features contributed significantly less, with Month (6.6%) and Crop-Code (4.5%) being the next most relevant. Temperature variables (Tmin, Tmax), Soil Moisture, and Rainfall value each contributed around 2–3%, indicating that the model relies primarily on cumulative rainfall, with limited influence from other factors.

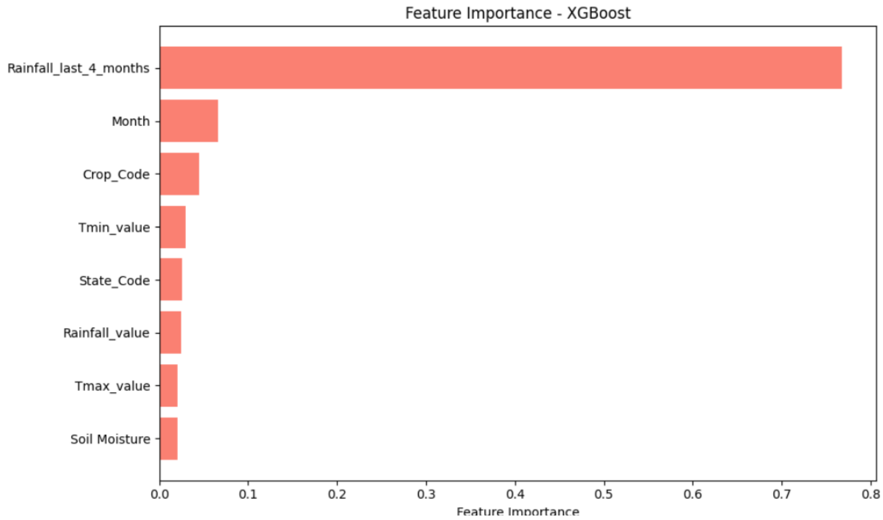

Figure 3.9: features of XGBOOST

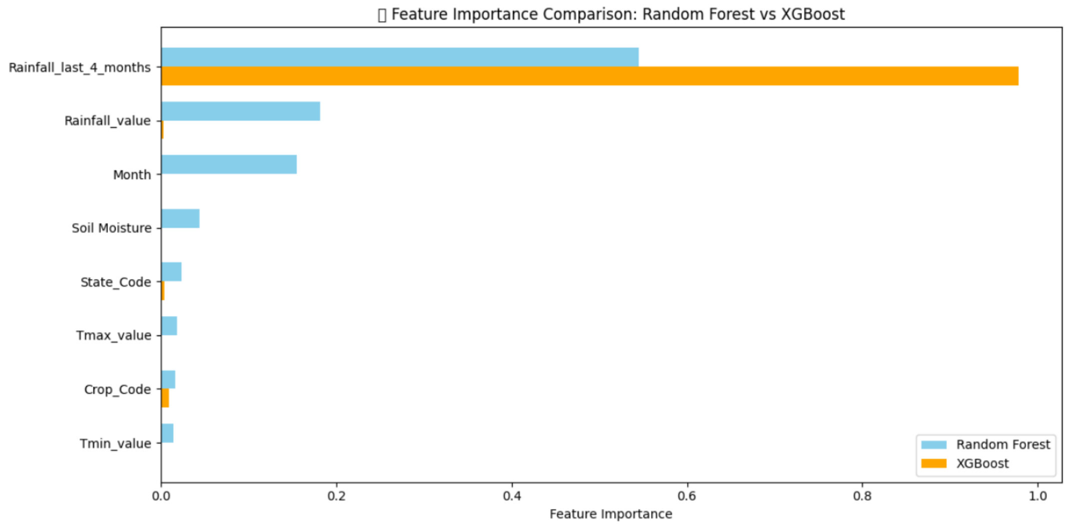

Figure 3.10: Feature importance XGBOOST vs Random Forest

Both models identified Rainfall-last-4-months as the most critical factor in determining irrigation need, aligning well with agronomic practices.

3.3.5 Threshold Tuning for F1 Score

Since the dataset was imbalanced, the default classification threshold of 0.5 may not have been the most optimal for achieving a balanced performance between precision and recall. To optimize the model's F1 score, a range of thresholds from 0.00 to 1.00 was tested.

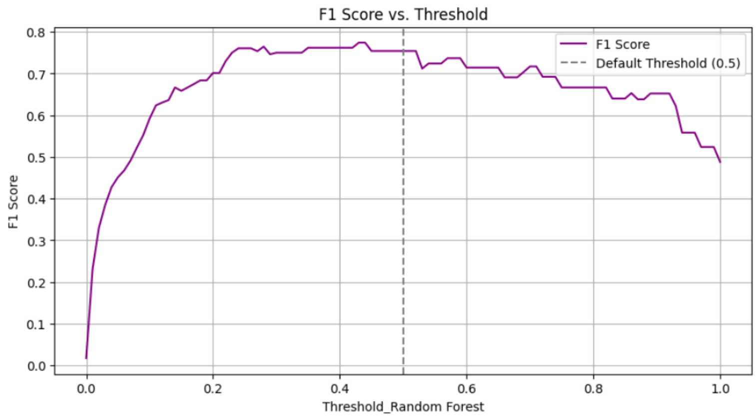

Figure 3.11: Feature importance XGBOOST vs Random Forest

- ▪ The best threshold was found to be 0.43, which improved the balance between precision and recall.
- ▪ The model was then re-evaluated using this new threshold, and the confusion matrix and classification report were regenerated to reflect the improved performance.

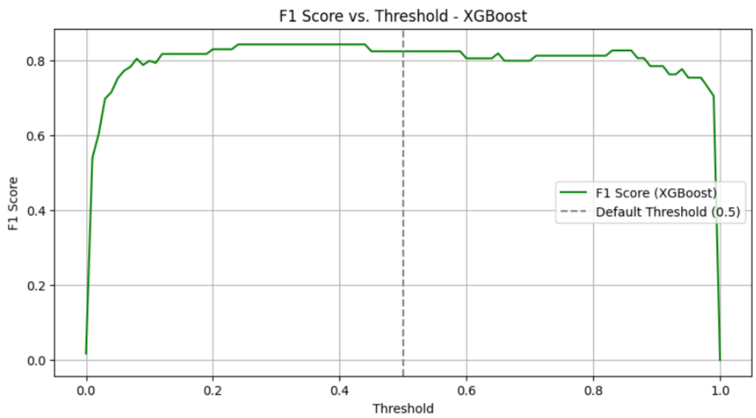

Figure 3.12: Feature importance XGBOOST vs Random Forest

3.3.6 Model Explainability with SHAP

To gain insights into the model's decision-making process, SHAP (SHapley Additive exPlanations) values were calculated. SHAP helps explain how each feature contributes to individual predictions, which is particularly important for understanding model behavior in applications like irrigation management.

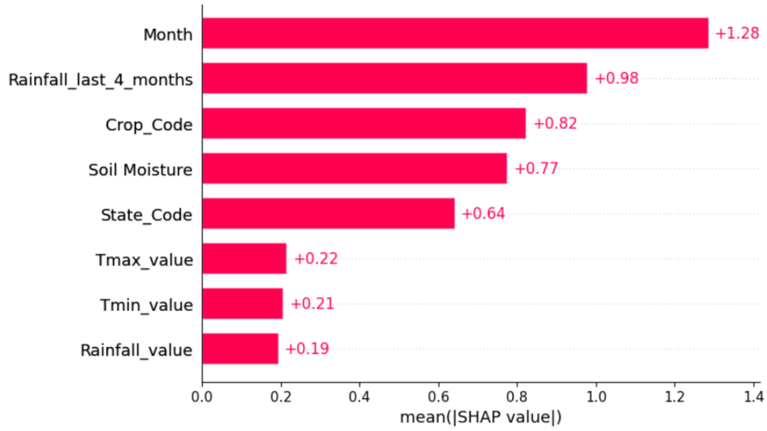

Figure 3.13: Feature importance XGBOOST vs Random Forest

- A SHAP explainer was initialized using the Random Forest model.
- SHAP values were computed for each feature and prediction, and visualized using a SHAP bar plot, which shows the average feature contribution.
- The focus was on explaining the model's decisions for class 1 (Irrigation Needed), helping stakeholders understand which factors, such as soil moisture and rainfall, most influence the decision to irrigate.

This approach improves the transparency of the model, making it easier for decision-makers to trust and interpret the results in the context of agricultural practices.

Chapter 4

Model Deployement

4.1 MLOps in AgroAI: Model Development to Deployment Pipeline

4.1.1 Model Training and Versioning

AgroAI leverages a suite of supervised learning models for its core features:

- **Crop Recommendation:** Models such as Logistic Regression, Decision Tree, and Random Forest were trained using agricultural datasets with features like soil nutrients (N, P, K), temperature, humidity, pH, and rainfall. Training was performed offline using `scikit-learn`, and models were serialized using `joblib` for deployment.
- **Smart Irrigation:** A Random Forest model was trained on state-wise weather data, including maximum and minimum temperature, rainfall, crop type, and soil moisture. Categorical features (state and crop) were encoded using label encoders.
- **Crop Health Monitoring:** A deep learning-based image classification model using GoogLeNet was fine-tuned on 18 classes of crop diseases. Training was conducted using `PyTorch Lightning`, with metrics tracked via `torchmetrics`. The model was trained on GPU-enabled hardware and exported using `state_dict` (.pth) for efficient CPU-based inference.

All models are version-controlled through serialized files (.pkl and .pth) to ensure traceability and reproducibility.

4.1.2 Data Preprocessing and Inference Pipeline

Each model has a corresponding preprocessing pipeline integrated within the Streamlit interface:

- **Crop Recommendation and Smart Irrigation:**
  - Users input agro-environmental data either manually or via APIs (e.g., OpenWeather).
  - The data is converted into a `pandas.DataFrame` and passed to the trained model.
  - Predictions are displayed in an intuitive UI with alerts based on outcomes.
- **Crop Health Monitoring:**
  - Users upload leaf images of crops.

- The images are preprocessed using the same pipeline as during training (resizing, normalization).
- The model returns a class index which is mapped to the corresponding disease label.

This approach ensures consistency between training and inference, aligning with MLOps principles.

4.1.3 Deployment with Streamlit

The full inference pipeline is deployed using `Streamlit`, a lightweight and rapid ML deployment tool. Key benefits include:

- **Modular Design:** Each model is activated based on user selection from the sidebar.
- **User Interface:** Designed for usability, featuring dynamic input fields and contextual prediction messages.
- **Image Handling:** The crop health module supports real-time image classification via browser UI.

This design facilitates rapid prototyping and feedback cycles, key aspects of MLOps and DevOps methodologies.

4.1.4 Monitoring and Maintenance

While advanced tools like MLflow or Prometheus were not integrated, several lightweight monitoring practices are in place:

- **CPU-based Inference:** A custom `CPUUnpickler` is used for loading GPU-trained models on CPU.
- **Error Handling:** Basic alerts are implemented for invalid API responses (e.g., incorrect city names).
- **Model Updates:** Modular model architecture allows swapping or upgrading without modifying core logic.

For future scalability, tools like Weights & Biases, MLflow, and FastAPI may be adopted for CI/CD, deployment automation, and real-time monitoring.

4.1.5 API Integration

AgroAI integrates with third-party APIs to fetch real-time environmental data:

- **OpenStreetMap (Nominatim):** Used to obtain location-specific coordinates.
- **OpenWeatherMap API:** Used to fetch real-time temperature and humidity data.

This integration simulates edge computing scenarios and demonstrates the fusion of machine learning and cloud services—an essential MLOps capability.

4.1.6 Conclusion

AgroAI embodies a functional yet lightweight MLOps framework tailored for academic and prototype-level applications. From model training and serialization to inference and user interaction, all components are designed to be modular, interpretable, and extensible. The use of `Streamlit` accelerates the model-to-market cycle, enabling real-time interaction with ML models.

Future improvements may include:

- Continuous training pipelines,
- Real-time monitoring dashboards,
- Cloud-native deployment using Docker and Kubernetes.

Chapter 5

Discussion and Analysis

5.1 Crop Recommendation System

The crop recommendation feature processes soil composition and environmental parameters to predict optimal crop selection:

Input Parameters:

Soil nutrients: Nitrogen (90), Phosphorus (42), Potassium (43) Environmental conditions: Temperature (40.98°C), Humidity (6Location data: Agra, Uttar Pradesh, India

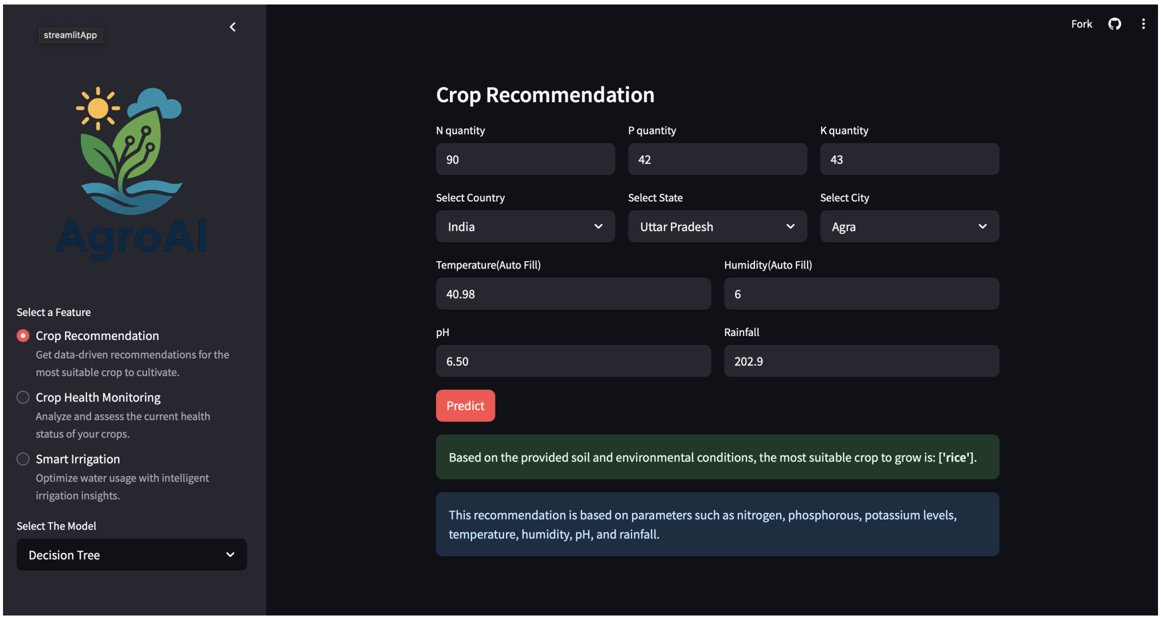

Figure 5.1: Crop Recommondation Dashboard

Prediction Mechanism:

The system likely employs a Decision Tree model (visible in the UI dropdown) These algorithms analyze relationships between input parameters and successful crop outcomes Historical data about crop performance under similar conditions informs recommendations The model evaluates which crops historically thrive with specific nutrient profiles and environmental conditions Location data helps account for regional growing patterns and climate trends

5.2 Disease Detection

This disease detection module uses image recognition to identify plant health issues:

Input Parameters:

Visual data: Uploaded images of plant foliage showing potential disease symptoms Image characteristics: Leaf coloration, spotting patterns, wilting, and other visual disease markers

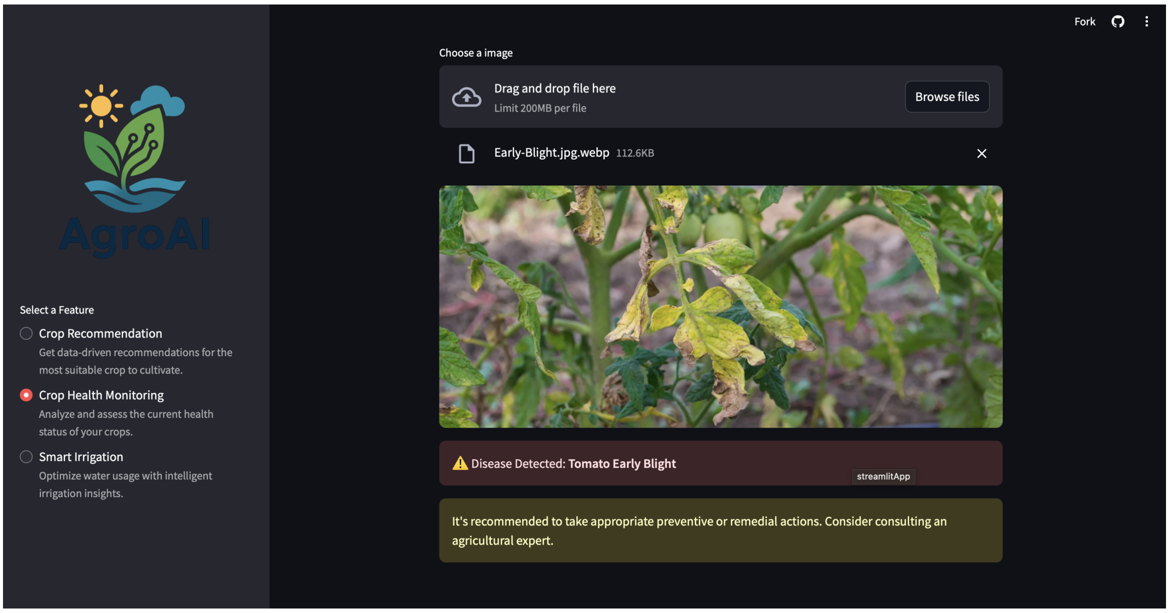

Figure 5.2: Disease Detection Dashboard

Prediction Mechanism:

Computer vision and image processing analyze plant appearance Pattern recognition algorithms compare uploaded images against a database of known disease presentations The system identifies specific conditions (e.g., "Tomato Early Blight") based on visual symptoms Early detection allows for timely intervention before diseases spread throughout crops Real-time diagnosis without requiring in-person expert consultation

5.3 Smart Irrigation System

The irrigation management module evaluates water needs based on multiple environmental factors:

Input Parameters:

Soil moisture level (0.0629) Temperature range (max: 39.29°C, min: 23.033°C) Rainfall metrics: Current (6.89 mm) and recent history (10.38 mm in last 7 days) Seasonal timing (Month: 04/April) Geographic location (Rajasthan) and crop type (Tomato)

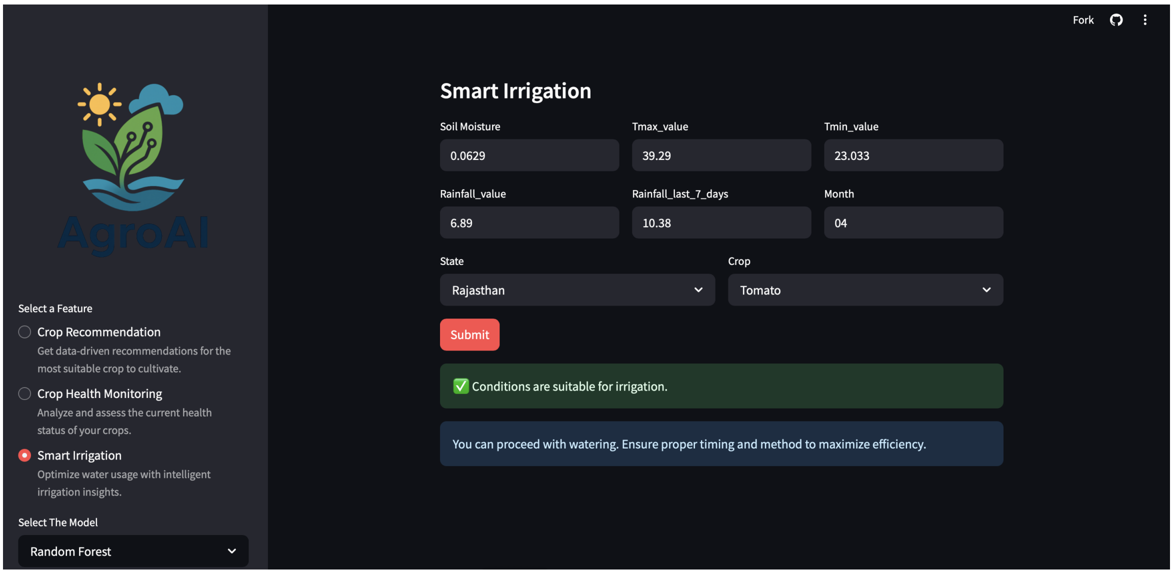

Figure 5.3: Smart Irrigation Dashboard

Prediction Mechanism:

Employs a Random Forest model (as indicated in the dropdown) Combines multiple decision trees to analyze complex relationships between variables Evaluates soil moisture against optimal levels for the specific crop Factors in recent precipitation to avoid overwatering Considers temperature trends that affect evaporation rates Adjusts recommendations based on crop-specific water requirements Provides binary outcomes (suitable/not suitable) for irrigation decisions

Chapter 6

Important Links and refrences

- **AgroAI Streamlit Page:** Visit Here
- **GitHub Repository** Visit Here
- **Crop Recommondation data**Visit Here
- **Dataset for Potato**Visit Here
- **Dataset for Tomato**Visit Here
- **Dataset for Rice**Visit Here
- **Dataset for Wheat**Visit Here
- **Dataset for Onion**Visit Here
- **Yearly Gridded Minimum Temperature** Visit Here
- **Yearly Gridded Maximum Temperature**Visit Here
- **Yearly Gridded Rainfall** Visit Here
- **Soil Moisture** Visit Here